Amrit Thapa, EIT

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Professional Summary

Mechanical Engineer (EIT) with hands-on experience in thermal-fluid systems, CFD/FEA simulation, and with exposure to semiconductor packaging and advanced manufacturing processes. Completed graduate-level coursework and applied project work in semiconductor packaging technologies, thermal interface solutions, and equipment design. Experienced in prototype development, experimental system validation, and cross-functional engineering collaboration. Seeking opportunities in energy systems, advanced manufacturing, and R&D- anywhere that demands both analytical depth and creative problem-solving.

Education

Master of Science in Mechanical Engineering *2022–2024*
The University of Texas at Tyler, TX GPA: 3.6/4.0
Bachelor of Engineering in Mechanical Engineering *2019–2021*
Tribhuvan University, Nepal GPA: 3.63/4.0

Relevant Training & Credentials

- **Semiconductor Packaging Manufacturing** – Arizona State University *2024*
- **Graduate Certificate: Thermal-Fluid Systems** – UT Tyler *2024*
- **Graduate Certificate: Systems & Controls** – UT Tyler *2024*
- **Engineer in Training (EIT)** – Texas Board of Professional Engineers *2023*
- **Introduction to Programming with MATLAB** – Vanderbilt University *2023*
- **CFD Workshop on OpenFOAM** – Kathmandu University *2020*

Technical Skills

- **Simulation & Analysis**: ANSYS (Fluent, CFX), OpenFOAM, MATLAB, CFD, FEA
- **Design & CAD**: SolidWorks, AutoCAD, CATIA
- **Thermal & Energy Systems**: OpenStudio, HAP, thermography, HVAC diagnostics
- **Programming & Data**: Python, MATLAB, Excel
- **Utility & Infrastructure Tools**: SPIDA CALC, KATAPULT, GIS, REVIT

Engineering Projects

Thermal Interface & Heat Transfer Optimization – Evaporative Condenser System
Performed thermal-fluid modeling and iterative design optimization of a condenser assembly, achieving a 24% reduction in compression work through enhanced heat transfer performance. Applied heat transfer analysis methods directly relevant to thermal management challenges in semiconductor packaging environments.

CFD-Driven Equipment Design – Closed-Circuit Subsonic Wind Tunnel
Led end-to-end design and simulation of a closed-circuit wind tunnel using ANSYS Fluent, optimizing flow uniformity and minimizing turbulence for precision test conditions. Co-developed axial fan specification using ANSYS CFX – mirroring equipment definition, selection, and validation workflows common in semiconductor tooling development.

Finite Element Analysis – Structural Stress Modeling (Nitsche's Method)
Developed a MATLAB-based FEM solver for stress analysis of complex geometries, improving solution stability and convergence. Demonstrated experimental design and computational validation methodology applicable to package-level structural reliability analysis.

Machine Learning for Thermal System Optimization – HVAC Controls
Built predictive Python models to optimize temperature control and improve energy efficiency in dynamic thermal systems, demonstrating ability to integrate data-driven methods into engineering workflows.

Temperature-Compensated MOSFET Voltage Reference Circuit
Designed and simulated a stable voltage reference circuit for variable environmental conditions, gaining exposure to electronics-level design considerations relevant to semiconductor packaging integration.

Work Experience

Engineer I – Techserv Engineering and Consulting, TX *2024–Present*

- Designed utility-pole configurations and produced PE-ready field packages, applying structural load analysis and mechanical design principles to real-world deployment constraints.
- Reviewed and validated stress reports using SPIDA Designer, ensuring NESC compliance and system integrity across large-scale infrastructure projects.
- Developed detailed engineering documentation to support field implementation and cross-functional coordination across multidisciplinary teams.
- Generated cost estimates and evaluated design trade-offs under tight project timelines, balancing performance, cost, and manufacturability.
- Collaborated with field teams to troubleshoot design and deployment issues, improving system reliability and process efficiency.

Engineering Intern – InergyX Building Solutions, TX *2023–2024*

- Conducted experimental testing (blower door and duct leakage) under PE supervision to evaluate system performance and air tightness – developing hands-on instrumentation and validation skills.
- Applied thermography and diagnostic tools to assess thermal behavior and identify system inefficiencies, recommending targeted HVAC improvements.
- Used drones and HVAC software for energy auditing; contributed to internal innovation projects and client-facing reporting.

Graduate Researcher – Entegra Sources Pvt. Ltd., Nepal *2021*

- Designed and fabricated a 5 tons/hr plastic recycling system (cutter and shredder assembly) using SolidWorks and AutoCAD, from concept through prototype.
- Led remote CFD and FEA simulations for diverse engineering projects, demonstrating independent technical execution across manufacturing applications.

Mechanical Design Intern – Surya Nepal Pvt. Ltd., Nepal *2019*

- Improved automated packaging line efficiency by 6% through geometric optimization and process redesign.
- Designed and fabricated a hydraulic press for tobacco-cake quality control, improving production consistency and repeatability.

Extracurricular Activities

Principal CAD Designer – Intercollegiate Go-Kart Project *2019*
Designed and developed a competition-ready go-kart prototype using CAD tools, managing design iteration from concept to physical build.